

FABRICATION OF DOUBLE-SIDED MICROFLUIDIC STRUCTURES VIA 3D PRINTED TRANSFER MOLDING

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Novelty / Progress Claims

We demonstrate single-step molding of two-sided PDMS devices with 3D printed molds for the first time. Advancing the 3D printed transfer molding technique from single-sided microfluidics¹ to multi-layer microfluidic manufacturing, this work utilizes the ease of 3D printing to create multiple molds with alignment structures to shape multiple layers of PDMS structures and quickly assemble them at the end. Using this single-step double-sided molding method, we demonstrate the molding of complex geometries in PDMS – including *vias*, built-in alignment marks, thin membranes, and rounded channels – that are difficult or impossible to achieve using standard soft lithography.

Background / State of the Art

The most far-reaching breakthrough in microfluidics has been the development of soft-lithography - using rigid micromachined molds to shape elastomeric polymers, commonly Poly(dimethylsiloxane) (PDMS), used for its low cost, strength, transparency, and especially biocompatibility.² Traditional methods for fabricating microfluidic devices use photolithography to construct micro-molds with; however, the process can be lengthy and costly. 3D printing presents a unique route to building complex microfluidic devices without requiring molding steps.³ Although 3D printed microfluidics is currently limited by: (1) printer resolution; (2) surface roughness; and (3) material types,⁴ the rapid developments of the 3D printing technologies are expected to address these matters. Additionally, while direct 3D printing is a rapid process for prototyping, for making multiple copies of microfluidic devices, it is slower, more expensive, and less reliable than transfer molding.

Description of the New Method

Figure 1 illustrates the fabrication techniques and capabilities of two-sided 3D printed transfer. (a) First, a mold is 3D Printed from a CAD model, cleaned of wax and then treated with a fluorinated silane (PFOTS, Sigma Aldrich) for hydrophobicity, after which the molds are fitted together using built-in mold alignment marks. (b) Next, the mold filled with PDMS and vacuum which is then cured. Excess PDMS is cut away using a scalpel, after which the mold is removed. (c) PDMS device has built-in inlets/outlets, membranes, and *vias*, and is (d) bonded to glass to create enclosed, arbitrary channels. The complex multi-level microfluidic device in (e) is fabricated using double-sided molding techniques and exhibits numerous design features, such as two layers of overlapping fluid flow, multiple microfluidic *vias*, built-in fluid inlets/outlets, as well as an elliptical 350 μ m domed membrane, a “Quake”-style membrane valve,⁵ and integrated alignment marks for use generating multi-layer devices.

Experimental Results:

Figure 2 shows conceptual illustrations of four different varieties of alignment marks which can be used to (a) properly align the molds for double-sided transfer molding; (b) to align a subsequent mold to PDMS or stably mount the PDMS; (c) align two layers of PDMS for assembly of a multi-layer microfluidic device; and (d) control the thickness of PDMS in situations where level curing is unreliable. **Figure 3** shows a range of features that can be generated by two-sided molding techniques including microfluidic *vias* to enable out-of-plane fluid detours, 250 μ m thick membranes, and fully rounded microfluidic channels composed of bonded complementary PDMS devices.

In **figure 4**, we demonstrate Membrane (or “Quake”) valve with a single molding step. The device consists of a microfluidic channel that is gated by a 600 μ m thick membrane valve. When pressure is applied to the gate, the membrane deforms, occluding fluid flow from the source to the drain (using MOSFET-analog terminology). **Figure 5** shows the characteristic curves of the membrane valve. Gate pressure (P_G) was increased smoothly and source pressure (P_S) was increased parametrically. Results showed approximately linear Q - P_G relationship within the range 160-200 kPa. P_S did not substantially affect initial Q drop.

Word Count 590

References:

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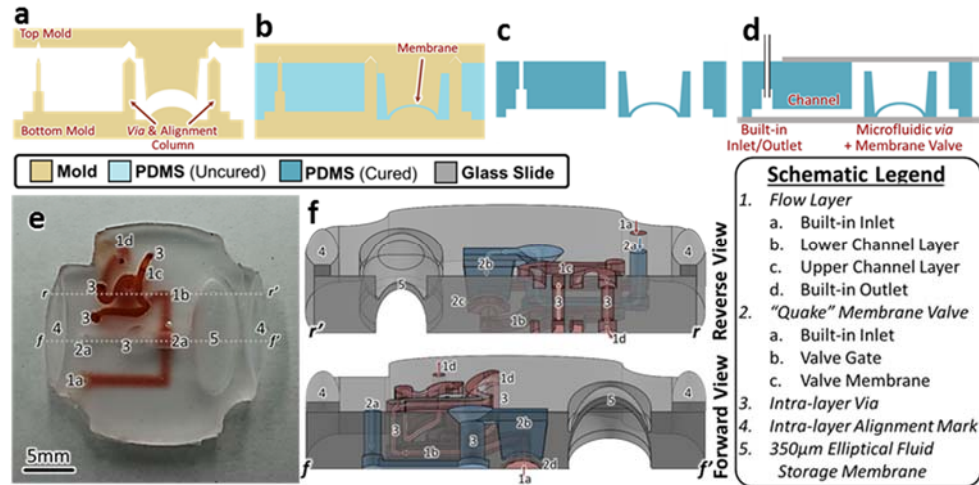


Figure 1: Illustration of the techniques and capabilities of 3D printed transfer molding for two-sided microfluidic devices. *Top:* fabrication techniques. (a) Mold is 3D Printed from CAD model, treated, and fitted using alignment marks, and (b) filled with PDMS and cured. Excess PDMS is cut away and the mold is removed. (c) PDMS device has built-in inlets/outlets, membranes, and *vias*, and is (d) bonded to glass to create enclosed, arbitrary channels. *Bottom:* technique capabilities. The multi-level microfluidic device in (e) is fabricated using double-sided molding techniques and exhibits design features, such as two layer fluid flow, multiple microfluidic *vias*, built-in fluid inlets/outlets, an elliptical 350µm domed membrane, and a “Quake”-style membrane valve, as well as alignment marks for use generating multi-layer devices.



Figure 2: Conceptual illustration of four different varieties of alignment marks: (a) Mold-Mold marks, used to manufacture PDMS molded on both sides; (b) Mold-PDMS marks, used for mounting microfluidic devices or for a stamping procedure; (c) PDMS-PDMS marks, used for bonding two complementary pieces of PDMS; and (d) PDMS Height Limiters, which use a glass slide to force PDMS into a specified thickness with a flat, uniform surface.

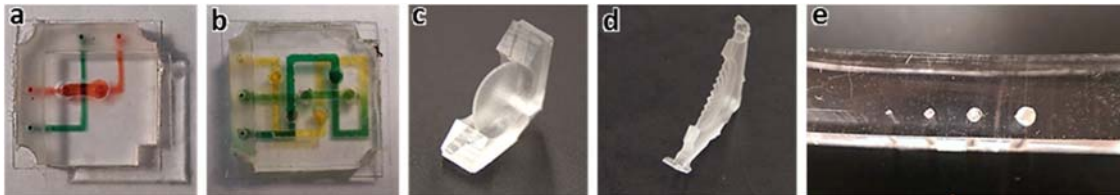


Figure 3: Features generated by two-sided transfer molding. (a,b) Microfluidic *vias* allow for more complex routing of fluid flow in 3D. (c,d) Thin membranes allow for membrane valves and fluid capacitors. (e) Fully circular channels fabricated by bonding two complementary devices using built-in PDMS-PDMS alignment marks.

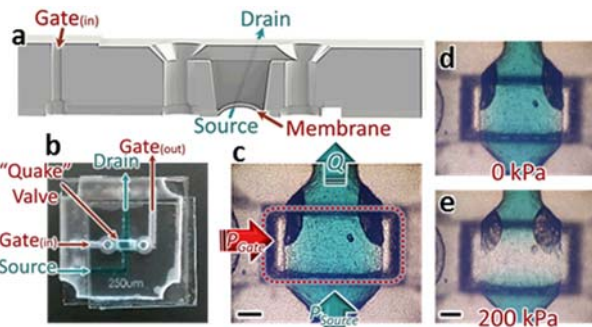


Figure 4: Membrane (“Quake”) valves generated by two-sided transfer molding. (a) Conceptual illustration showing membrane valve, source, drain, and gate. (b) Top-down photo of device. (c) Microscope image showing the active valve region and key components. (d) Active membrane region and full closure deformation under 200 kPa. Scale bar = 500µm.

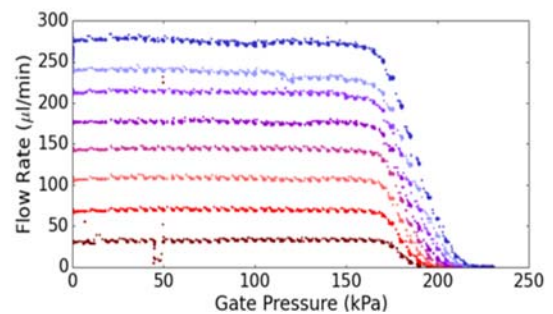


Figure 5: Results of membrane valve parametric pressure sweep. Gate pressure was gradually increased until Source-Drain flow cut off, and then Source pressure was increased. Results show ~linear Gate Pressure-Flow Rate relationship within a window of 160-200 kPa. Pressure of initial flow reduction did not change substantially with source pressure.